MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 05 (SOLUTIONS)
Chapter 3: Pair of Linear Equations in Two Variables

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions
MATHS MANTHAN | (3 × 1 = 3 Marks)

Q1. For what value of k do the equations $3x - y + 8 = 0$ and $6x - ky = -16$ represent coincident lines? [1]

(A) 1/2 (B) −1/2 (C) 2 (D) −2

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 2** [0.5 Mark]

Rewriting second equation: $6x - ky + 16 = 0$. For coincident lines:

Formula / Theorem Used: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Justification: $\frac{3}{6} = \frac{-1}{-k} = \frac{8}{16} \implies \frac{1}{2} = \frac{1}{k} \implies \mathbf{k = 2}$. [0.5 Mark]

Q2. 8 chairs and 5 tables cost Rs. 10500, while 5 chairs and 3 tables cost Rs. 6450. The cost of each chair will be: [1]

(A) Rs. 750 (B) Rs. 600 (C) Rs. 850 (D) Rs. 900

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) Rs. 750** [0.5 Mark]

Justification: Let chair = Rs. x, table = Rs. y: $8x + 5y = 10500$ and $5x + 3y = 6450$. [0.5 Mark]

Multiplying by 3 and 5: $24x + 15y = 31500$ and $25x + 15y = 32250 \implies x =$ Rs. **750**.

Q3. If the sum of the ages of a father and his son is 65 years and twice the difference of their ages is 50 years, then father's age is: [1]

(A) 40 years (B) 45 years (C) 55 years (D) 65 years

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 45 years** [0.5 Mark]

Justification: Let father's age = x, son's age = y. $x + y = 65$ and $2(x - y) = 50 \implies x - y = 25$. [0.5 Mark]

Adding: $2x = 90 \implies x =$ **45** years.

SECTION B: Assertion-Reason Question — MATHS MANTHAN | (1 × 1 = 1 Mark)

Q4. **Assertion (A):** If a pair of linear equations represents coincident lines, then the system has infinitely many solutions.
Reason (R): Two coincident lines overlap completely and have every point on the line in common. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: Every point common to two plotted lines is a solution of the system; since coincident lines share all infinitely many points, both (A) and (R) are true and (R) explains (A). [0.5 Mark]

SECTION C: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q5. Find the coordinates of the points where the line $2x - 3y = 6$ intersects the x-axis and the y-axis. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Formula / Theorem Used: On x-axis, $y = 0$; on y-axis, $x = 0$.

Step 1: x-axis Intersection ($y = 0$) [1 Mark]

Putting $y = 0$ in $2x - 3y = 6$: $2x - 3(0) = 6 \implies 2x = 6 \implies x = 3$. Point: $(3, 0)$.

Step 2: y-axis Intersection ($x = 0$) [1 Mark]

Putting $x = 0$ in $2x - 3y = 6$: $2(0) - 3y = 6 \implies -3y = 6 \implies y = -2$. Point: $(0, -2)$.

x-axis: $\mathbf{(3, 0)}$, y-axis: $\mathbf{(0, -2)}$

Q6. Solve the following pair of linear equations for x and y:

$$x + y = a + b \quad \text{and} \quad ax - by = a^2 - b^2$$

[2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $x + y = a + b$ — (i) and $ax - by = a^2 - b^2$ — (ii)

Step 1: Eliminating y [1 Mark]

Multiplying equation (i) by b: $bx + by = ab + b^2$ — (iii)

Adding (ii) and (iii): $(a + b)x = (a^2 - b^2) + (ab + b^2) = a^2 + ab = a(a + b)$.

Assuming $a + b \neq 0$, dividing both sides by $(a + b)$ gives $x = a$.

Step 2: Solving for y [1 Mark]

Substituting $x = a$ into (i): $a + y = a + b \implies y = b$.

$\mathbf{x = a, \quad y = b}$

Q7. A fraction becomes $\frac{9}{11}$, if 2 is added to both the numerator and the denominator. If 3 is added to both the numerator and the denominator, it becomes $\frac{5}{6}$. Find the fraction. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let the fraction be $\frac{x}{y}$ $(y \neq 0)$.

Step 1: Setting Up Both Linear Equations [1.5 Marks]

$\frac{x+2}{y+2} = \frac{9}{11} \implies 11(x+2) = 9(y+2) \implies 11x - 9y = 18 - 22 = -4$ — (i)

$\frac{x+3}{y+3} = \frac{5}{6} \implies 6(x+3) = 5(y+3) \implies 6x - 5y = 15 - 18 = -3$ — (ii)

Step 2: Solving by Substitution [1.5 Marks]

From (i), $x = \frac{9y-4}{11}$. Substituting into (ii):

$$6\left(\frac{9y-4}{11}\right) - 5y = -3 \implies 54y - 24 - 55y = -33 \implies -y = -9 \implies y = 9$$

Then $x = \frac{9(9)-4}{11} = \frac{77}{11} = 7$.

$$\boxed{\text{Required Fraction } = \mathbf{\frac{7}{9}}}$$

Q8. Under railway reservation rules, the railway half ticket costs half the full basic fare, but the reservation charge is the same on a half ticket as on a full ticket. One reserved first class ticket from station A to station B costs Rs. 2530. Also, one reserved first class ticket and one reserved first class half ticket from A to B costs Rs. 3810. Find the full first class fare from station A to B, and also the reservation charge for a ticket. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let full first class basic fare = Rs. x and fixed reservation charge = Rs. y.

Formula / Theorem Used: Full Ticket $= x + y$, Half Ticket $= \frac{x}{2} + y$

Step 1: Forming Equations [2 Marks]

Condition I (1 full ticket): $x + y = 2530$ — (i)

Condition II (1 full + 1 half ticket): $(x+y)+\left(\frac{x}{2}+y\right) = 3810 \implies \frac{3}{2}x+2y = 3810 \implies 3x+4y = 7620$ — (ii)

Step 2: Solving by Elimination [2 Marks]

Multiplying equation (i) by 3: $3x + 3y = 7590$ — (iii)

Subtracting (iii) from (ii): $(3x + 4y) - (3x + 3y) = 7620 - 7590 \implies y = 30$.

Step 3: Finding Full Fare x [1 Mark]

From (i): $x + 30 = 2530 \implies x = 2530 - 30 = 2500$.

$$\boxed{\text{Full first class fare } = \text{Rs. } \mathbf{2{,}500}, \qquad \text{Reservation charge } = \text{Rs. } \mathbf{30}}$$

Q9. CASE STUDY: Wildlife Sanctuary Census & Ecological Survey

During an educational excursion to Nandan Kannan Wildlife Sanctuary, students of Class X visited the bird sanctuary and deer park. Rohan, a keen observer, put forward a riddle to his classmates regarding the animal population. The forest ranger confirmed that the combined population of birds and deer had exactly 1000 eyes and 1400 legs. Let x denote the number of two-legged birds and y denote the number of four-legged deer. [4]

(i) Write down the linear equation representing the total count of eyes. [1]

(ii) Write down the linear equation representing the total count of legs. [1]

(iii) How many birds (x) and how many deer (y) are there in the sanctuary? [1]

(iv) Find the ratio of the number of birds to the number of deer in the sanctuary. [1]

— OR —

Find the total number of animals (birds + deer) in the sanctuary. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Biological Attributes: Each bird (x) has 2 eyes and 2 legs; each deer (y) has 2 eyes and 4 legs.

Formula / Theorem Used: Total Eyes $= 2x + 2y = 1000$ and Total Legs $= 2x + 4y = 1400$

Sub-part (i): Linear Equation for Eyes [1 Mark]

$\boxed{\mathbf{2x + 2y = 1000}}$ (or simplified: $x + y = 500$) — (i)

Sub-part (ii): Linear Equation for Legs [1 Mark]

$\boxed{\mathbf{2x + 4y = 1400}}$ (or simplified: $x + 2y = 700$) — (ii)

Sub-part (iii): Finding x (Birds) and y (Deer) [1 Mark]

Subtracting $2x + 2y = 1000$ from $2x + 4y = 1400$: $2y = 400 \implies y = 200$.

Substituting in $x + y = 500$: $x = 500 - 200 = 300$. $\boxed{\mathbf{x = 300}}$ birds, $\boxed{\mathbf{y = 200}}$ deer

Sub-part (iv): Ratio of Birds to Deer [1 Mark]

$x : y = 300 : 200 = \boxed{\mathbf{3 : 2}}$.

Sub-part (iv) [OR Option]: Total Number of Animals [1 Mark]

Total Animals $= x + y = 300 + 200 = \boxed{\mathbf{500}}$ animals.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★